
Comparison methods for semilinear elliptic problems on Riemannian manifolds with a Ricci lower bound

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[**Revisión Estado.**] Versión matemáticamente reparada, 29 de septiembre de 2026. Azul: texto nuevo o reformulado. Rojo: motivo, cambio de alcance y pendientes. Fuente recibido y primera auditoría preservados en `audits/2026-09-29/`. A se formula bajo criterio de perfil; B bajo máximo suave de una cara; C tiene hipótesis y prueba nuevas. Esta revisión requiere lectura de los autores antes de enviarse.

Abstract

We study positive solutions of a semilinear Dirichlet problem on bounded domains in a Riemannian manifold with a Ricci lower bound. A sign condition on a chosen one-dimensional profile, together with a comparison of boundary slopes, yields a pointwise gradient bound. Equality determines a normalized warped-product metric on the domain. We verify the profile condition for the affine family $f(u) = nku + c$ and an explicit non-linear Euclidean family, and show by a separate example that the inequality $f' \geq nk$ alone does not imply it. Under a smooth maximum-set hypothesis we derive area and volume comparisons. For $f(u) \geq nku + 1$, we also obtain a normalized lower bound for the distance of the maximum set from the boundary, with rigidity for space-form balls.

Keywords: Semilinear elliptic equations, gradient comparison, Ricci curvature, maximum sets, hot spots.

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1 Introduction

In elliptic PDE one often encounters a tight interplay between the geometry of a domain and the analytic features encoded by a solution. Global scales such as the inradius and the volume, together with local boundary information, constrain the maximum value of the solution, its maximum set and the behaviour of the gradient along level sets.

Let $(\mathcal{M}^n, g)$ be a complete Riemannian manifold, $n \geq 2$, satisfying

$$\text{Ric}_g \geq (n - 1)kg. \quad (1.1)$$

We consider a bounded connected domain $\Omega \Subset \mathcal{M}$ with C^2 boundary and a positive solution of

$$\Delta u + f(u) = 0 \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega. \quad (1.2)$$

Write $M = \max_{\overline{\Omega}} u$ and $\text{Max}(u) = \{u = M\}$. The comparison is performed separately on each connected component $\mathcal{U}$ of $\Omega \setminus \text{Max}(u)$.

Gradient comparison with one-dimensional models has a substantial history. Kröger [Krö92] and Bakry–Qian [BQ00] developed spectral comparison methods under Ricci bounds; Valtorta [Val12] and Naber–Valtorta [NV14] treated the p -Laplacian. For semilinear equations, Farina–Valdinoci [FV10] prove gradient estimates in Euclidean domains with nonnegative boundary mean curvature. Their compact-manifold result [FV11] concerns a manifold without boundary and a different gradient bound.

Andrews–Xiong [AX19] use inverse model profiles and two-point functions. Their Theorem 27 assumes nonnegative Ricci curvature and mean-convex boundary. For a negative Ricci lower bound κ , their Theorem 30 requires $H_{\partial\Omega} \geq (n - 1)\sqrt{-\kappa}$ in their convention. Their work also includes rigidity results. Related semilinear rigidity theorems based on P -functions appear in [CFP25].

Our formulation follows the decomposition along the maximum set used by Agostiniani–Borghini–Mazzieri [ABM25], its higher-dimensional version [ABBM26], and the spherical comparisons in [EM26b, EM26a]. In each component, we invert one monotone branch of a radial ODE profile. The boundary slope of that profile must dominate the slope of the given solution on the corresponding wall. A second, explicit profile condition ensures that the weighted gradient difference satisfies an elliptic inequality with the sign needed by the maximum principle.

Theorem A gives the comparison and its intrinsic equality case. Proposition 2.1 verifies the profile condition for $f(u) = nku + c$, $c > 0$, on its positive range. Example 2.2 gives a nonlinear family with $f' > 0$ and $f'' < 0$ in the Euclidean case. Theorem B is a volume–maximum-area inequality under a smooth maximum-set hypothesis. Theorem C gives the normalized distance estimate. The criterion is imposed on the chosen model; it is not asserted to follow from $f' \geq nk$ for an arbitrary nonlinearity.

[Revisión Estructural.] Se retiran del texto activo los teoremas generales no justificados sobre rangos y monotonía de $\bar{r}$, el argumento de deslizamiento, el límite hiperbólico incorrecto y las figuras que dependían de ellos. No se borra ese material: se conserva el fuente original completo. La versión actual concentra la contribución en una cadena A–B–C demostrada bajo hipótesis explícitas. Recuperar mayor generalidad es trabajo posterior, no una afirmación de esta versión.

1.1 Radial notation

For $0 < r < \bar{r}_k$, where $\bar{r}_k = \pi/\sqrt{k}$ if $k > 0$ and $\bar{r}_k = +\infty$ otherwise, set

$$s_k(r) = \begin{cases} \sin(\sqrt{k}r)/\sqrt{k}, & k > 0, \\ r, & k = 0, \\ \sinh(\sqrt{-k}r)/\sqrt{-k}, & k < 0, \end{cases} \quad \cot_k(r) = \frac{s'_k(r)}{s_k(r)}, \quad \tan_k(r) = \frac{s_k(r)}{s'_k(r)}.$$

The last expression is used only where its denominator is positive. A model branch is a solution $\bar{U}$ of

$$\bar{U}'' + (n-1) \cot_k \bar{U}' + f(\bar{U}) = 0 \tag{1.3}$$

with maximum M at its core radius $\bar{R}$, strictly monotone between the core and a finite zero r_b . The outer branch has $r_b = \bar{r}_+ > \bar{R}$; the inner branch has $0 < r_b = \bar{r}_- < \bar{R}$. A centered branch has $\bar{R} = 0$ and is outer. The inverse branch $\chi : [0, M] \rightarrow I_b$ is continuous, with $\chi(0) = r_b$ and $\chi(M) = \bar{R}$. Here I_b is the closed interval between the wall and the core. We use the same symbol for its interior when integrating.

The profile equation is independent of a choice of fiber metric. Its geometric realization on a positive-radius interval is $dr^2 + s_k(r)^2 g_F$. A centered model has a smooth pole only when g_F is the unit round metric. In an equality case the fiber metric is determined from the actual wall by normalization, as in Theorem A. We do not fix an unnormalized boundary metric before making that identification.

2 Gradient comparison

Let Ω be a bounded $\mathcal{C}^2$ domain in a complete Riemannian manifold with $\text{Ric} \geq (n-1)kg$. Let $u \in \mathcal{C}^\infty(\Omega) \cap \mathcal{C}^2(\bar{\Omega})$ solve $\Delta u + f(u) = 0$, $u > 0$ in Ω , $u = 0$ on $\partial\Omega$, where $M = \max u$, f is smooth on a neighbourhood of $[0, M]$, and $f > 0$ on $(0, M]$. Fix a monotone branch of a radial profile $\bar{U}$, solving

$$\bar{U}'' + (n-1)\cot_k(r)\bar{U}' + f(\bar{U}) = 0, \quad \bar{U}(\bar{R}) = M, \quad \bar{U}'(\bar{R}) = 0,$$

with a finite zero r_b and $\bar{U}' \neq 0$ between the core and the wall. For a centered branch, use the regular solution at $\bar{R} = 0$. All positive radii under consideration belong to $(0, \bar{r}_k)$. Let χ be the inverse branch and $\Psi = \chi \circ u$. For r in the open branch set

$$\lambda(r) = \frac{-\bar{U}''(r) + \cot_k(r)\bar{U}'(r)}{\bar{U}'(r)^2}, \quad \mu(r) = f'(\bar{U}(r)) - nk + \frac{n+2}{n}\lambda(r)f(\bar{U}(r)).$$

We call this branch gradient-admissible when $\mu \geq 0$ throughout the branch. This is a condition on the chosen profile, in addition to its existence and monotonicity. The inequality $f' \geq nk$ alone does not imply it. The closure of every component $\mathcal{U}$ meets $\partial\Omega$: otherwise u would attain an interior minimum below M , contradicting $\Delta u = -f(u) < 0$.

[Revisión A.] Sustituida la condición falsa $f' \geq nk \Rightarrow \mu \geq 0$ por la hipótesis explícita $\mu \geq 0$ sobre el perfil elegido. La prueba se rehace con peso positivo, deriva corregida y fibra normalizada. El alcance es menor que el enunciado original; la proposición afín de abajo demuestra que incluye una familia completa, no una condición vacía.

Theorem A (Gradient comparison under the profile criterion). *Let $\mathcal{U}$ be a connected component of $\Omega \setminus \text{Max}(u)$. Suppose the chosen profile is gradient-admissible and*

$$\max_{\partial\Omega \cap \bar{\mathcal{U}}} |\nabla u|^2 \leq \bar{U}'(r_b)^2.$$

Then

$$|\nabla u|^2 \leq \bar{U}'(\Psi)^2 \quad \text{in } \mathcal{U}.$$

If equality holds at an interior point, then $\text{Ric}(\nabla u, \nabla u) = (n-1)k|\nabla u|^2$ and $\mathcal{U}$ is intrinsically isometric to a warped product with metric

$$g = dr^2 + \left(\frac{s_k(r)}{s_k(r_b)} \right)^2 g_{\Gamma_b},$$

where $\Gamma_b = \partial\Omega \cap \bar{\mathcal{U}}$, and $u = \bar{U}(r)$. In particular Γ_b is connected. In the centered case the smooth completion at the maximum is a geodesic ball in the space form of curvature k . No assertion about the metric outside Ω is made.

Proof. Set $W = |\nabla u|^2$, $q = \bar{U}'(\Psi)^2$, $a = 2(n-1)/n$ and

$$\beta = \left(\frac{s_k(\Psi)}{|\bar{U}'(\Psi)|} \right)^a, \quad F = \beta(W - q).$$

The identity proved below gives

$$\Delta F - \frac{2(n-2)}{n} \lambda(\Psi) \langle \nabla F, \nabla u \rangle - \frac{2(n-1)}{n} \frac{W}{q} \mu(\Psi) F \geq 0.$$

On the wall $F \leq 0$ by the assumed slope comparison. Near the maximum set, Taylor's theorem applied to the nonnegative function $M - u$ gives $W = O(M - u)$. If $\bar{R} > 0$, then $q = 2f(M)(M - u) + O((M - u)^{3/2})$, hence $F = O((M - u)^{1/n}) \rightarrow 0$. In the centered case, $\bar{U}'(r) = -f(M)r/n + o(r)$ and $s_k(r) = r + o(r)$, so β is bounded and $F \rightarrow 0$ as well. Thus F extends continuously to $\bar{\mathcal{U}}$, with nonpositive boundary values. The maximum principle, or its application to the relatively compact set $\{F > \varepsilon\}$, gives $F \leq 0$.

If F vanishes at an interior point, the strong maximum principle gives $F \equiv 0$. Equality in the defect identity implies

$$\nabla^2 u = -\lambda du \otimes du + \frac{\lambda W - f(u)}{n} g, \quad \text{Ric}(\nabla u, \nabla u) = (n-1)kW.$$

Since $W = q > 0$ in $\mathcal{U}$, the chain rule gives

$$|\nabla \Psi| = 1, \quad \nabla^2 \Psi = \cot_k(\Psi)(g - d\Psi \otimes d\Psi).$$

For $0 < a_1 < b_1 < M$, the set $\{a_1 \leq u \leq b_1\} \cap \bar{\mathcal{U}}$ is compact, so this unit-speed flow exists between any two interior levels. Hopf's lemma extends the identification to a collar of the wall. Integrating the flow of $\nabla \Psi$ therefore identifies all regular levels and gives the stated metric, normalized at $r = r_b$. Connectedness of $\mathcal{U}$ implies connectedness of each level. In the centered case smoothness at the pole forces the normalized link to be the unit round sphere, proving the intrinsic ball conclusion. $\square$

Proposition 2.1 (A nonempty explicit class). *Let $f(x) = nkx + c$, with $c > 0$, and suppose $f > 0$ on $[0, M]$. Every monotone admissible radial branch with maximum M is gradient-admissible. For $\bar{R} > 0$ one has $\mu > 0$; for the centered branch one has $\lambda = \mu = 0$.*

Proof. Put $h = \bar{U}'/s_k$. Differentiating the radial equation and using $s_k'' = -ks_k$ and $\cot_k' = -k - \cot_k^2$ yields

$$h'' + (n+1) \cot_k h' = 0.$$

For $\bar{R} > 0$, $h'(\bar{R}) = -f(M)/s_k(\bar{R})$, and therefore

$$h'(r) = -\frac{f(M)s_k(\bar{R})^n}{s_k(r)^{n+1}}, \quad \lambda(r) = \frac{f(M)s_k(\bar{R})^n}{s_k(r)^n \bar{U}'(r)^2} > 0.$$

Since $f' = nk$, the assertion follows from the definition of μ . For a centered regular profile, $s_k^{n+1}h'$ is constant and its limit at zero is zero. Hence $h' = 0$, so $\lambda = \mu = 0$. $\square$

2.1 A nonlinear admissible family

Example 2.2. Let $k = 0$, $n \geq 2$ and $0 < \varepsilon < n/(2(n+2))$. Put $M = 1 - \varepsilon$ and

$$\bar{U}(r) = M - r^2 + \varepsilon r^4, \quad 0 \leq r \leq 1, \quad f(t) = 2(n+2)\sqrt{1 - 4\varepsilon(M-t)} - 4.$$

Then f is smooth and positive on a neighbourhood of $[0, M]$, and this centered profile is gradient-admissible. Indeed, writing $d(r) = 1 - 2\varepsilon r^2 > 0$, we have

$$\bar{U}' = -2rd, \quad f(\bar{U}) = 2n - 4(n+2)\varepsilon r^2 = -\bar{U}'' - \frac{n-1}{r}\bar{U}'.$$

The profile decreases from M to zero at $r = 1$. The radicand in f is bounded below by $(1 - 2\varepsilon)^2 > 0$ on $[0, M]$, while $f(0) = 2n - 4(n+2)\varepsilon > 0$. Direct calculation gives

$$f'(\bar{U}) = \frac{4(n+2)\varepsilon}{d}, \quad \lambda = -\frac{2\varepsilon}{d^2}, \quad \mu = \frac{16(n+2)\varepsilon^2 r^2}{nd^2} > 0 \quad (0 < r \leq 1).$$

In particular this is a nonlinear, increasing, strictly concave family: $f''(t) = -8(n+2)\varepsilon^2/[1 - 4\varepsilon(M-t)]^{3/2} < 0$. If additionally $\varepsilon \leq (2n-1)/(4(n+2))$, then $f \geq 1$ on $[0, M]$, as required in Theorem C for $k = 0$. Applying either comparison theorem to another domain still requires the corresponding boundary-slope domination by this profile. Here f and the maximum value $M = 1 - \varepsilon$ are linked; the calculation certifies the displayed centered branch, not every branch of the same f .

[Revisión Ejemplo no lineal.] Familia explícita nueva, verificada por cálculo exacto y lectura independiente. Amplía la clase de perfiles comprobados más allá del caso afín. No demuestra que la concavidad de una función arbitraria baste, ni resuelve la selección automática de pendientes.

2.2 Why the derivative condition alone is insufficient

Example 2.3. Take $n = 3$, $k = 0$, $f(x) = e^x + e^{2x}$ and

$$\bar{U}(r) = \log \frac{2}{1+r^2}, \quad 0 \leq r \leq 1.$$

This profile has maximum $M = \log 2$, vanishes at $r = 1$, and satisfies (1.3). Indeed, with $q_0 = 1 + r^2$,

$$\bar{U}' = -\frac{2r}{q_0}, \quad \bar{U}'' = \frac{2(r^2-1)}{q_0^2}, \quad f(\bar{U}) = \frac{2}{q_0} + \frac{4}{q_0^2}.$$

Although $f' > 0$ and $f(x) \geq f(0) = 2$ for $x \geq 0$, direct substitution gives

$$\lambda = -1, \quad \mu = -\frac{4r^2}{3(1+r^2)^2} < 0 \quad (0 < r \leq 1).$$

Thus the inequality $f' \geq nk$ does not imply the profile criterion. This example refutes that implication; it does not by itself assert that every gradient comparison outside the criterion fails.

[Revisión Contraejemplo.] Cálculo analítico comprobado independientemente. Se incorpora al artículo para delimitar con precisión la hipótesis nueva. El experimento con anillos y $f = e^u$ queda en la auditoría como evidencia numérica, sin atribuirle certificación rigurosa.

2.3 The exact defect identity

Write a dot for differentiation with respect to u and evaluate λ, μ at Ψ . Let

$$T = \nabla^2 u + \lambda du \otimes du + \frac{f(u) - \lambda W}{n} g.$$

Then

$$\begin{aligned} \Delta F - \frac{2(n-2)}{n} \lambda \langle \nabla F, \nabla u \rangle - \frac{2(n-1)}{n} \frac{W}{q} \mu F &= 2\beta(|T|^2 + \text{Ric}(\nabla u, \nabla u) \\ &\quad - (n-1)kW). \end{aligned} \quad (2.1)$$

Indeed, the radial equation gives

$$\dot{q} = 2\bar{U}'', \quad \frac{\dot{\beta}}{\beta} = a\lambda, \quad \dot{\lambda} + (1-a)\lambda^2 = \frac{\mu}{q}.$$

Expand

$$|T|^2 = |\nabla^2 u|^2 + \lambda \langle \nabla W, \nabla u \rangle + \frac{n-1}{n} \lambda^2 W^2 + \frac{2}{n} f(u) \lambda W - \frac{f(u)^2}{n}.$$

Insert this expression in Bochner's identity and use $\Delta q = \ddot{q}W - \dot{q}f(u)$. Writing $D = W - q$, the terms independent of the curvature defect become

$$\Delta D + 2\lambda \langle \nabla D, \nabla u \rangle = -a\lambda^2 WD + a\lambda f(u)D + 2|T|^2 + 2(\text{Ric}(\nabla u, \nabla u) - (n-1)kW).$$

Multiplication by β , followed by the two identities for its derivative and for $\dot{\lambda}$, gives (2.1). The Ricci bound and $\mu \geq 0$ are now exactly the signs required in the proof.

[Revisión B.] Se sustituye el máximo meramente Lipschitz por una hipersuperficie compacta C^3 , con una sola cara de la componente. Se demuestra el límite de flujo mediante gráficas normales uniformes. El caso Lipschitz general queda retirado de esta versión; permanece documentado como extensión abierta. El campo correcto lleva s_k^{n-1} .

3 Area and volume comparison

For the following result assume additionally that $u \in \mathcal{C}^{\max\{n,3\}}(\Omega)$, so that Sard's theorem applies, and that $\Gamma_M = \bar{\mathcal{U}} \cap \text{Max}(u)$ is a nonempty compact embedded $\mathcal{C}^3$ hypersurface without boundary, and that $\mathcal{U}$ occupies exactly one local side of each component of Γ_M . This explicit hypothesis avoids unproved convergence claims at singular strata or multiple-sheet limits. Assume $\bar{R} > 0$.

Proposition 3.1 (Area comparison under the smooth maximum hypothesis). *Under the hypotheses of Theorem A and the smooth maximum hypothesis above, put $\Gamma_t = \{u = t\} \cap \bar{\mathcal{U}}$ for a regular value $t \in [0, M)$. Then*

$$\frac{\mathcal{H}^{n-1}(\Gamma_t)}{s_k(\chi(t))^{n-1}} \geq \frac{\mathcal{H}^{n-1}(\Gamma_M)}{s_k(\bar{R})^{n-1}}.$$

Equality for one such value holds if and only if the intrinsic warped-product rigidity conclusion of Theorem A holds.

Proof. Let

$$X = \frac{\nabla u}{s_k(\Psi)^{n-1} \bar{U}'(\Psi)}, \quad \text{div } X = \frac{f(u)(W - q)}{s_k(\Psi)^{n-1} \bar{U}'(\Psi)^3}.$$

For regular levels define

$$J(t) = \int_{\Gamma_t} \frac{|\nabla u|}{s_k(\chi(t))^{n-1} |\bar{U}'(\chi(t))|} d\sigma.$$

The divergence theorem on $\{t < u < M - \varepsilon\} \cap \mathcal{U}$ and the sign $W - q \leq 0$ imply $J(t) \geq J(M - \varepsilon)$ on either branch. Also $J(t) \leq \mathcal{H}^{n-1}(\Gamma_t)/s_k(\chi(t))^{n-1}$.

The Hessian of u annihilates tangent vectors to Γ_M because $\nabla u = 0$ there. Its normal eigenvalue is $-f(M) < 0$. In one-sided normal coordinates (y, s) along this compact hypersurface,

$$u(y, s) = M - \frac{1}{2}f(M)s^2 + O(s^3), \quad |\nabla u(y, s)|^2 = f(M)^2 s^2 + O(s^3),$$

uniformly in y . Thus for small $\varepsilon > 0$ the level $M - \varepsilon$ is a single normal graph over each component of Γ_M , with $s = \sqrt{2\varepsilon/f(M)} + O(\varepsilon)$. There are no other high-level components, by compactness and the definition of Γ_M . The graph metrics converge to the metric of Γ_M and $|\nabla u|/|\bar{U}'(\Psi)| \rightarrow 1$ uniformly. Consequently

$$\lim_{\varepsilon \downarrow 0} J(M - \varepsilon) = \frac{\mathcal{H}^{n-1}(\Gamma_M)}{s_k(\bar{R})^{n-1}}.$$

This proves the estimate. Equality forces $J(t)$ to equal the limiting flux at the maximum. Monotonicity then makes $J(s)$ constant for regular $s \in (t, M)$. Applying the divergence theorem between two such interior levels shows that the integral of the nonnegative gradient deficit is zero.

Since $f > 0$ and all remaining factors are nonzero inside that slab, $W = q$ there. Thus equality holds at an interior point, even when $t = 0$, and Theorem A gives rigidity. The converse follows directly from the warped-product area element. $\square$

Theorem B (Volume–maximum-area comparison). *Under the hypotheses of the preceding proposition, let I_b be the interval between the chosen wall r_b and the core $\bar{R}$. Then*

$$\frac{\mathcal{H}^n(\mathcal{U})}{\mathcal{H}^{n-1}(\Gamma_M)} \geq \frac{1}{s_k(\bar{R})^{n-1}} \int_{I_b} s_k(r)^{n-1} dr.$$

Equality holds if and only if the intrinsic warped-product rigidity conclusion of Theorem A holds.

Proof. The critical set of u has zero n -dimensional measure. Indeed, at each critical point $\text{tr } \nabla^2 u = -f(u) < 0$, so some component of ∇u has a nonzero derivative. Its zero set is locally a hypersurface containing the critical set; use a countable cover. The coarea formula and the preceding two comparisons therefore give

$$\begin{aligned} \mathcal{H}^n(\mathcal{U}) &= \int_0^M \int_{\Gamma_t} \frac{1}{|\nabla u|} d\sigma dt \\ &\geq \int_0^M \frac{\mathcal{H}^{n-1}(\Gamma_t)}{|\bar{U}'(\chi(t))|} dt \\ &\geq \frac{\mathcal{H}^{n-1}(\Gamma_M)}{s_k(\bar{R})^{n-1}} \int_0^M \frac{s_k(\chi(t))^{n-1}}{|\bar{U}'(\chi(t))|} dt. \end{aligned}$$

The substitution $t = \bar{U}(r)$ identifies the last integral with $\int_{I_b} s_k(r)^{n-1} dr$. If equality holds, the nonnegative gradient-comparison deficit vanishes almost everywhere at regular points; continuity gives equality at an interior regular point. Theorem A applies. Conversely, the warped-product metric gives equality directly. $\square$

4 Distance from the maximum set to the boundary

[Revisión C.] Se corrige $f \leq nku + 1$ a $f \geq nku + 1$. Se demuestra el principio del máximo mediante el primer autovalor, se usa la pendiente global para todas las componentes y se explicitan una vecindad normal fuertemente convexa y los rangos de k y M . El caso de igualdad determina sólo el dominio y los valores de f alcanzados por la solución. La desigualdad es estricta para todo núcleo positivo.

Theorem C (Location of the maximum set). *Let $(\mathcal{M}^n, g)$ be a complete Riemannian manifold, $n \geq 2$, with $\text{Ric}_g \geq (n-1)kg$, and let $\Omega \Subset \mathcal{M}$ be a connected C^2 domain contained in a strongly convex normal neighbourhood. In particular, for every $p \in \Omega$, the distance $r_p =$*

$\text{dist}(p, \cdot)$ is smooth on the punctured inscribed ball $B_{d(p)}(p) \setminus \{p\}$, where $d(p) = \text{dist}(p, \partial\Omega)$. Let u be a positive classical solution of $\Delta u + f(u) = 0$ in Ω , with $u = 0$ on $\partial\Omega$, and put $M = \max_{\bar{\Omega}} u$ and $r_\Omega = \max_{\bar{\Omega}} d$. Assume the regularity and positivity hypotheses of Section 2, and

$$f(t) \geq nkt + 1 \quad (0 \leq t \leq M).$$

Fix a branch of a model profile $\bar{U}$ with maximum M , core radius $\bar{R}$, and boundary radius $\bar{r}_\diamond$, where $\diamond \in \{+, -\}$. Write $\chi^\diamond$ for its inverse and assume

$$\max_{\partial\Omega} |\nabla u|^2 \leq \bar{U}'(\bar{r}_\diamond)^2, \quad \mu(r) = f'(\bar{U}(r)) - nk + \frac{n+2}{n} \lambda(r) f(\bar{U}(r)) \geq 0$$

on the chosen open branch, with $\lambda = (-\bar{U}'' + \cot_k \bar{U}')/\bar{U}'^2$. If $k > 0$, assume $r_\Omega < \pi/(2\sqrt{k})$; if $k < 0$, assume $M < 1/(n|k|)$. Set

$$\ell_\diamond = \begin{cases} \bar{r}_+ - \bar{R}, & \diamond = +, \\ \bar{R} - \bar{r}_-, & \diamond = -. \end{cases}$$

Then, for every $p \in \text{Max}(u)$,

$$\frac{n d(p)}{\tan_k(r_\Omega)} \geq \frac{\ell_\diamond}{\sqrt{2M/n + kM^2}}.$$

Equality holds if and only if the chosen model is centred, $\bar{\Omega}$ is isometric to the closed ball of radius r_Ω in the simply connected space form of curvature k , and, under this isometry,

$$u(x) = W_{r_\Omega}(\text{dist}(o, x)), \quad f(t) = nkt + 1 \quad (0 \leq t \leq M),$$

where

$$W_R(r) = \begin{cases} \frac{s'_k(r)/s'_k(R) - 1}{nk}, & k \neq 0, \\ \frac{R^2 - r^2}{2n}, & k = 0. \end{cases}$$

In the equality case $\text{Max}(u) = \{o\}$. In particular, the inequality is strict for every model with $\bar{R} > 0$, including every inner branch.

Proof. We first justify the linear comparison used below. Let $\phi_1 > 0$ be an eigenfunction for the first Dirichlet eigenvalue $\lambda_1(\Omega)$. Green's formula gives

$$(\lambda_1(\Omega) - nk) \int_\Omega u \phi_1 = \int_\Omega (f(u) - nku) \phi_1 \geq \int_\Omega \phi_1 > 0.$$

Consequently $\lambda_1(\Omega) > nk$. The operator $L = -\Delta - nk$ therefore satisfies the Dirichlet maximum principle on Ω and on every subdomain of Ω .

Fix $p \in \Omega$ and set $R = d(p)$. The function W_R is positive on $[0, R)$, vanishes at R , and satisfies

$$W_R'' + (n-1) \cot_k(r) W_R' + nk W_R = -1, \quad W_R'(r) = -\frac{s_k(r)}{ns_k'(R)} < 0 \quad (r > 0).$$

The radial function $w = W_R \circ r_p$ is smooth at its centre. Laplacian comparison and the sign of W_R' imply

$$Lw \leq 1 \quad \text{in } B_R(p), \quad Lu = f(u) - nk u \geq 1.$$

Since $u \geq w = 0$ on $\partial B_R(p)$, the maximum principle gives $u \geq w$ in $B_R(p)$. Thus

$$u(p) \geq h_k(d(p)), \quad h_k(R) := W_R(0) = \frac{2s_k(R/2)^2}{ns_k'(R)}.$$

The formula at $k = 0$ is understood as $h_0(R) = R^2/(2n)$.

Choose an incenter o of Ω . The function $q_k(t) = 2t/n + kt^2$ is strictly increasing on $[0, M]$ under the stated assumptions. The identity

$$q_k(h_k(R)) = \frac{\tan_k(R)^2}{n^2}$$

and $h_k(r_\Omega) \leq u(o) \leq M$ yield

$$\frac{\tan_k(r_\Omega)}{n} \leq \sqrt{q_k(M)}.$$

The same chosen profile is a comparison profile for every connected component of $\Omega \setminus \text{Max}(u)$, because its boundary slope dominates the maximum on all of $\partial\Omega$. Theorem A, applied on each component, therefore gives

$$|\nabla \Psi| \leq 1, \quad \Psi = \chi^\diamond \circ u,$$

away from $\text{Max}(u)$. Extend Ψ to the maximum set by $\Psi = \bar{R}$. This extension is locally Lipschitz: near the maximum, $|\chi^\diamond(t) - \bar{R}|$ is a smooth function of $\sqrt{M-t}$ which vanishes linearly at zero, and the square root of the nonnegative C^2 function $M-u$ is locally Lipschitz. Its weak gradient vanishes almost everywhere on the set where it is constant. Hence $|\nabla \Psi| \leq 1$ almost everywhere on Ω .

For $p \in \text{Max}(u)$, take a minimizing geodesic from p to a nearest point of $\partial\Omega$. Before its endpoint the geodesic lies in Ω . Integrating the Lipschitz bound along this geodesic gives

$$\ell_\diamond = |\bar{r}_\diamond - \bar{R}| \leq d(p).$$

Combining the last two inequalities proves the assertion.

Suppose equality holds. Both inequalities used in the last step must then be equalities, so $d(p) = \ell_\diamond$ and $h_k(r_\Omega) = u(o) = M$. The function $u - W_{r_\Omega} \circ r_o$ is nonnegative on $B_{r_\Omega}(o)$,

satisfies $Lv \geq 0$, and vanishes at o . The strong maximum principle gives $v \equiv 0$ on this ball. By continuity, u vanishes on its boundary. Positivity of u and connectedness of Ω imply $\Omega = B_{r_\Omega}(o)$. The inequalities $Lu \geq 1$ and $Lw \leq 1$ are now equalities, so $f(u) = nku + 1$ and

$$\Delta r_o = (n-1) \cot_k(r_o)$$

on the punctured ball. The equality case follows directly from Bochner's identity for r_o :

$$0 = |\nabla^2 r_o|^2 + \text{Ric}(\nabla r_o, \nabla r_o) + \langle \nabla r_o, \nabla \Delta r_o \rangle.$$

The Ricci bound and Cauchy–Schwarz force $\nabla^2 r_o = \cot_k(r_o)(g - dr_o^2)$. Integrating the induced metrics of the distance spheres, with their round limit at the smooth centre, gives $g = dr_o^2 + s_k(r_o)^2 g_{\mathbb{S}^{n-1}}$ on the ball. The radial profile is strictly decreasing, so its maximum is uniquely attained at o , and its range shows that $f(t) = nkt + 1$ on $[0, M]$.

Finally, if $\bar{R} > 0$, then near o the ball solution has

$$|\nabla u|^2 \sim \frac{2f(M)}{n}(M - u), \quad \bar{U}'(\Psi)^2 \sim 2f(M)(M - u).$$

Thus $|\nabla \Psi| \rightarrow 1/\sqrt{n} < 1$. Integrating along a radial minimizing segment gives $\ell_\diamond < r_\Omega = d(o)$, contradicting equality. Hence $\bar{R} = 0$ and the chosen branch is centred. Conversely, the centred space-form ball profile has $\mu = 0$, $\ell_+ = r_\Omega$, and $q_k(M) = \tan_k(r_\Omega)^2/n^2$, so equality holds. $\square$

A Existence of the radial profiles

[Revisión ODE.] Se sustituye el punto fijo inconsistente por una contracción de Volterra. Se prueba un criterio suficiente de ceros finitos: $f > 0$ en todo $[0, M]$. No se reivindican las antiguas afirmaciones generales de selección automática ni monotonía de pendientes. Los teoremas principales requieren un perfil elegido que exista y cumpla su criterio.

Proposition A.1. *Let f be smooth on a neighbourhood of $[0, M]$ and satisfy $f > 0$ on $[0, M]$. For every $\bar{R} \in [0, \bar{r}_k)$ there is a radial solution with $U(\bar{R}) = M$, $U'(\bar{R}) = 0$. If $\bar{R} > 0$, it has finite first zeros $0 < r_- < \bar{R} < r_+ < \bar{r}_k$, and is strictly monotone on each side of its maximum while positive. If $\bar{R} = 0$, the regular centered solution has a finite first zero $r_+ < \bar{r}_k$ and is strictly decreasing on $(0, r_+]$.*

Proof. For positive core radius, local existence and uniqueness follow from the regular ODE. At the center, use the Volterra operator

$$(\mathcal{TV})(r) = M - \int_0^r s_k(y)^{1-n} \int_0^y s_k(x)^{n-1} f(V(x)) dx dy.$$

On a closed supremum-norm ball about M in $C([0, \varepsilon])$, the estimate $s_k(r) = r + O(r^3)$ gives

$$\|\mathcal{T}V - M\|_\infty \leq C\varepsilon^2 \sup |f|, \quad \|\mathcal{T}V - \mathcal{T}Z\|_\infty \leq CL\varepsilon^2 \|V - Z\|_\infty,$$

where L is a local Lipschitz constant. Small ε makes the operator a contraction of that ball into itself. Its fixed point satisfies $U(0) = M$, $U'(0) = 0$ and $U''(0) = -f(M)/n$; differentiation for $r > 0$ gives the radial ODE. Smoothness and the usual even extension at the center follow successively from the integral equation.

Until the first zero, the flux formula is

$$U'(r) = -s_k(r)^{1-n} \int_{\bar{R}}^r s_k(t)^{n-1} f(U(t)) dt. \quad (\text{A.1})$$

It gives strict monotonicity on each side of $\bar{R}$. Let $m = \min_{[0, M]} f > 0$. A positive solution cannot stop at an interior radius of $(0, \bar{r}_k)$ while its value stays in $[0, M]$, since the flux and the regular ODE allow continuation there.

For an inner branch, the integral in (A.1) is bounded away from zero in absolute value near zero. Since $\int_0^\delta s_k(r)^{1-n} dr = +\infty$ for $n \geq 2$, positivity down to the origin would contradict $U \leq M$. Thus $r_- > 0$. For $k > 0$, the analogous divergence at $\bar{r}_k$ forces a finite outer zero. For $k = 0$, the bound $f(U) \geq m$ in the flux formula gives a negative derivative of order r for large r . For $k < 0$ it gives

$$\limsup_{r \rightarrow \infty} U'(r) \leq -\frac{m}{(n-1)\sqrt{-k}} < 0$$

if the solution were positive for all large radii. Both conclusions contradict positivity for all r , proving the finite outer zero. $\square$

Remark A.2. *The stronger requirement $f(0) > 0$ in Proposition A.1 is a sufficient existence condition, not a standing condition in Theorem A. Profiles with $f(0) = 0$ may also be used if their existence, finite wall and monotonicity are verified separately. Existence of profiles does not by itself ensure that their wall slopes dominate those of a given solution.*

Proposition A.3 (Energy ordering for nonpositive curvature). *If $k \leq 0$, $\bar{R} > 0$ and U is a positive two-sided profile, set $G(t) = \int_0^t f(s) ds$. Then*

$$\frac{U'(r_+)^2}{2} < G(M) < \frac{U'(r_-)^2}{2}.$$

Proof. The energy $E = U'^2/2 + G(U)$ satisfies $E' = -(n-1)\cot_k(r)U'^2$. Here $\cot_k > 0$ and U' vanishes only at the core. Therefore E strictly decreases from the inner wall to the outer wall, and $E(\bar{R}) = G(M)$. $\square$

B Scope of the revised comparison

The model criterion makes the maximum-principle argument explicit and verifiable. It is sufficient, and no necessity claim is made. It would be useful to characterize further nonlinearities for which every admissible branch satisfies it. The Lipschitz maximum-set case also requires a separate treatment of top-level fluxes and their multiplicities. Finally, the normal-neighbourhood assumption in the distance theorem may be relaxed using a weak Laplacian comparison across the cut locus; that extension is not used here.

Data availability

The results in this version are analytic. The computational illustrations of the earlier draft are not used in the proofs presented here.

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